Note

In C++, any **non zero is treated as true including negative numbers and zero is treated as false.**

Selection statements and iteration statements are executed depending upon the conditional expression. The conditional expression evaluates either true or false.

10.4 Selection statements

In a program a decision causes a one time jump to a different part of a program. Decisions in C++ are made in several ways, most importantly with if .. else … statement which chooses between two alternatives. Another decision statement, switch creates branches for multiple alternatives sections of code, depending on the value of a single variable.

10.4.1 if statement

The if statement evaluates a condition, if the condition is true then a true-block (a statement or set of statements) is executed, otherwise the true-block is skipped.The general syntax of the if statement is:

```
if (expression)
        true-block;
statement-x;
```

In the above syntax, **if** is a keyword that should contain expression or condition which is enclosed within parentheses. If the expression is true (nonzero) then the true-block is executed and followed by statement-x are also executed, otherwise, the control passes to statement-x. The true-block may consists of a single statement, a compound statement or empty statement. The control flow of **if** statement and the corresponding flow chart is shown below.

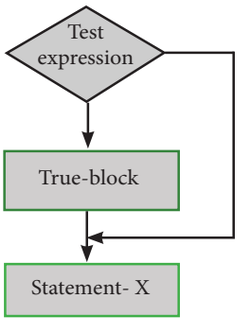

Illustration 10.1 C++ program to check whether a person is eligible to vote using if statement

```
#include <iostream>
using namespace std;
int main()
{
        int age;
        cout<< "\n Enter your age: ";
        cin>> age;
        if(age>=18)
                cout<< "\n You are eligible for voting ....";
        cout<< "This statement is always executed.";
        return 0;
}
```

The pair of braces is not required because if condition followed by only one statement

Output

```
Enter your age: 23
You are eligible for voting….
This statement is always executed.
```


Chapter 10 Page 152-179.indd 154 3/24/2020 9:21:32 AM